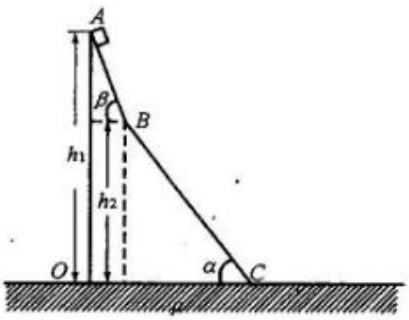
3. Wooden wedge $OABC$ of mass $M = 30.0$ kg rests on level ground. AB has slope $\beta = 60°$, BC has $\alpha = 45°$. AB and BC join at B in a tiny smooth arc. A small block $m = 4.00$ kg, treated as a point. Released from rest at A, $h_1 = 2.80$ m above ground, it slides down the frictionless slope. Maximum static and sliding ground friction coefficients $\mu = 6.00 \times 10^{-2}$. Table height of B: $h_2 = 2.00$ m. Neglect both speed changes at B. Find the work on the wedge as the block slides from A to the base C. Take $g = 10.0\ \mathrm{m \cdot s^{-2}}$.



4. (25 points)

Here E is source emf (negligible internal resistance); R: resistance; S: switch. Right of A、B, 8 identical ideal capacitors, each C. From closing S to full charge, find the heat in R.
In your solution, mark on the diagram the assumed sign of charge on every capacitor plate.

